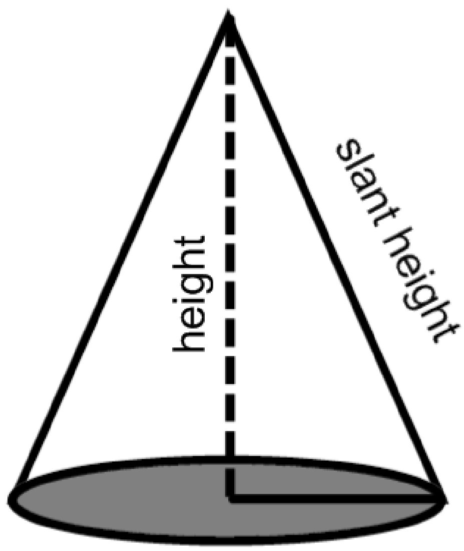
**Problem 2**

# Problem 2

**Problem 1** *A right cone has a fixed slant height (see figure below) of 9 ft. The cone's height is shrinking at a rate of 0.5 ft/sec. At what rate is the* ***volume*** *of the cone changing when the height is 6 ft? Be sure to label the picture.*



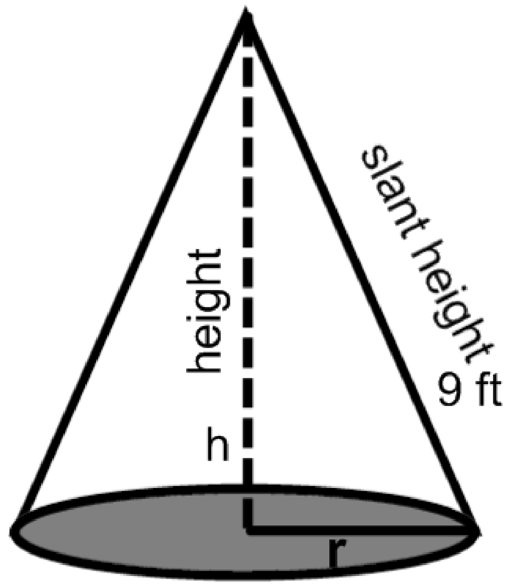

**Explanation.** We introduce the variables $V$, for the volume, $r$ for the radius

and $h$ for the height of the cone.

The given rate is $\frac{dh}{dt} = -0.5$ ft/s;

the rate to be determined is $\left[\frac{dV}{dt}\right]_{h=6}$.

We label the picture.




The equation relating all relevant variables is

$$V = \frac{1}{3}\pi r^2 \cdot h$$

Before we differentiate, we will express $r^2$ in terms of $h$.

$$r^2 = 81 - h^2$$

and we substitute this into the equation above

$$V = \frac{1}{3}\pi(81 - h^2) \cdot h$$

Then we simplify it.

$$V = \frac{1}{3}\pi(81h - h^3)$$

Now, we differentiate with respect to $t$.

$$\frac{dV}{dt} = \frac{\pi}{3}(81 - 3h^2) \cdot \frac{dh}{dt}$$

Now, we evaluate.

$$\left[\frac{dV}{dt}\right]_{h=6} = \frac{\pi}{3}(81 - 3(6)^2)(-0.5)$$
$$\left[\frac{dV}{dt}\right]_{h=6} = \frac{\pi}{3}(-27)(-0.5)$$
$$\left[\frac{dV}{dt}\right]_{h=6} = 4.5\pi \text{ ft}^3/\text{s}$$